

Week 6 – Workshop

Magnetics: Anomalies and Susceptibility



Geophysics 3A: Potential Fields & Geothermics

July 31, 2026

Preparation

Pre-reading: Course Notes, Ch. 8

Lecture videos:

Notes:

Purpose

This workshop builds physical intuition for magnetic fields and their geological sources. Students first build up dipole field geometry from direct observation, then apply that intuition to predict anomaly shapes for idealized structures, before turning to what actually carries magnetization in real rocks and how thermal structure limits it at depth.

Learning Outcomes

By the end of this workshop, students should be able to:

- Describe and predict magnetic field patterns generated by dipoles and combinations of sources
- Predict magnetic anomaly shapes from simple geological geometries, and contrast them with gravity anomalies for the same structures
- Relate magnetic susceptibility to rock type and geological process, and distinguish primary from secondary controls on it
- Explain the concept of Curie depth and its relationship to thermal structure
- Link source depth to anomaly wavelength and spectral content

Session Flow (?? min)

0–5 min: Warm-Up: Cutting a Magnet in Half

Thought experiment: if you cut a bar magnet exactly in half, does one piece become a pure “north” and the other a pure “south”?

- What do you predict happens to each half?
- If you kept cutting smaller and smaller pieces, would you ever reach a piece with only one pole?

5–40 min: Activity 6.A: Visualizing Magnetic Fields

- Iron filings with a single disc magnet, vertical and horizontal orientation
- Predict-observe-compare-reflect: two dipoles side by side
- A line of stacked magnets — field pattern near one end vs. along the side
- Optional extension: cube magnets, and whether shape alone can remove dipolar behavior at range

40–55 min: Mini-Lecture: Magnetic Fields

- Formalizing vocabulary: dipole vs. monopole, why isolated magnetic poles cannot exist
- Dipole convergence figures: two opposite point poles at decreasing separation, showing how a monopole-like appearance at close range gives way to a dipole pattern at distance
- How magnetism differs from gravity (vector nature, polarity)
- Preview: applying this field geometry to predict anomalies from real geological structures

55–80 min: Activity 6.B: Magnetic Structures

Like gravity, we want to gain intuition about the shapes of anomalies we may expect to see from various geologic features. In this activity, you will:

- Produce reasoning-based sketches of magnetic anomalies for idealized geometries
- Direct comparison with gravity anomalies for the same bodies
- Vertical vs. horizontal inducing field

80–90 min: Break**90–100 min: Warm-Up: Which Property Matters Most?**

Suppose you could know only one property of every rock in the crust. Which would best help predict magnetic anomalies: FeO concentration, magnetite abundance, magnetic susceptibility, density, remanent magnetization, or oxidation state?

Target idea: There isn't a perfect answer — each property captures part of the picture but misses something else. Today's activity will show why susceptibility depends on a trace mineral phase rather than on any single bulk property.

100–130 min: Activity 6.C: Magnetic Susceptibility of Rocks

- Station 1: reaction pairs (peridotite/serpentinite, basalt/altered basalt, eclogite) — magnetite created or destroyed by reaction
- Station 2: igneous suite — magnetite set by primary crystallization, no reaction involved
- Station 3: ore minerals — decoupling metal content from magnetism

130–160 min: Activity 6.D: Curie Depth and Magnetic Crustal Thickness

- Curie temperature and magnetic crustal thickness
- Depth–wavelength relationships
- Antarctica case study
- Back-of-the-envelope heat-flow estimates

160–170 min: Wrap-up

- What magnetism can and cannot tell us
- Connection to upcoming Fourier/spectral methods

170–180 min: Preview: Fourier Transforms and Potential Fields

Both gravity and magnetic anomalies contain information about source depth encoded in their spatial wavelengths: deep sources produce broad, long-wavelength anomalies while shallow sources produce narrow, short-wavelength anomalies. Next week we formalize this depth–wavelength relationship using Fourier transforms. We will see how gravity and magnetic data can be filtered, upward-continued, and reduced-to-the-pole in the wavenumber domain, and how sampling decisions control what geological features can be resolved.

Workshop Notes

Workshop Notes

Activity 6.A

Visualizing Magnetic Fields

Purpose

This activity develops physical intuition for magnetic fields as vector fields generated by dipoles. You will explore how field geometry depends on source orientation, on the interaction between multiple sources, and on how far you are from the source.

Learning Outcomes

By the end of this activity, you should be able to:

- Describe and sketch the field pattern of a single magnetic dipole in vertical and horizontal orientations.
- Predict and compare the field pattern produced by two interacting dipole sources.
- Identify how the field pattern near the end of a stacked line of magnets differs from the pattern along its side, and relate this to a finite geological structure.
- Explain why source orientation, not source strength alone, determines the shape of a magnetic anomaly.

Background

Iron filings align with the local field direction, so the pattern they form shows you field-line direction and curvature — where lines converge, diverge, and close. What filings do *not* give you is a reliable measure of field strength: filing density is too easily affected by how much you sprinkle and how the paper is tapped to treat as data. Throughout this activity, base your answers on field *shape and symmetry*, not on how densely the filings appear to cluster.

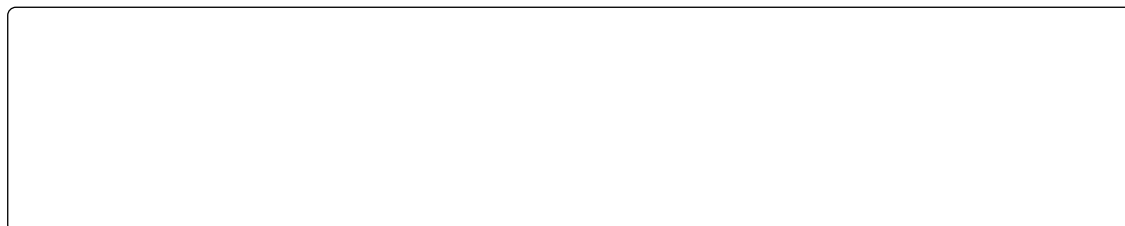
Materials

- Magnets
- Paper sheets
- Iron filings

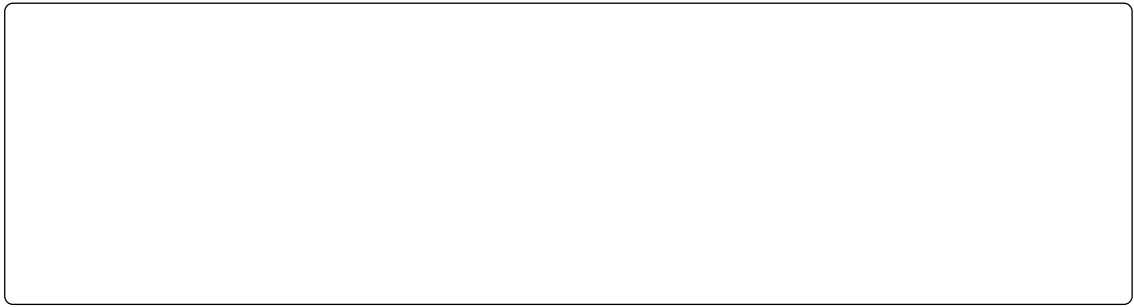
Question 1. Single magnet — vertical orientation

Place a disc magnet flat on the table and sprinkle iron filings over a sheet of paper on top.

1. Describe the symmetry of the field pattern.
2. Where are the field gradients strongest?
3. If this magnet were buried at depth, where would you expect the strongest anomaly?

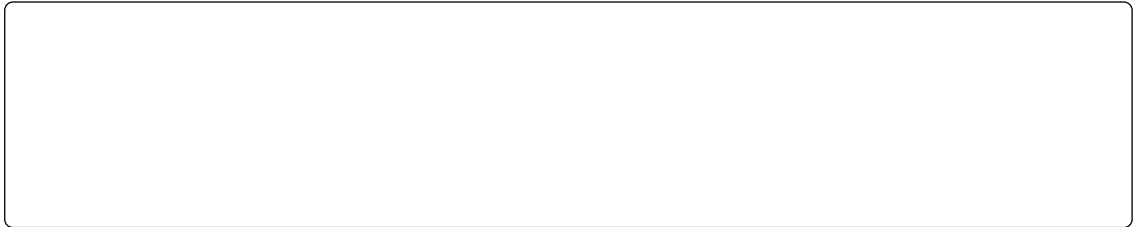


Sketch:

**Question 2.** Single magnet — horizontal orientation

Stand the same magnet on its edge and repeat the experiment.

1. How does the field symmetry differ from Part A?
2. Where are positive and negative lobes located?
3. How would this differ from a gravity anomaly produced by the same body?



Sketch:

**Question 3.** Two Dipoles Side by Side

Take two magnets magnets, oriented the same way (e.g. both vertical, both N up), and place them a short distance apart on the table.

Predict: Before sprinkling filings, sketch what you expect the combined field pattern to look like, based on what you saw for a single dipole in Part A.



Observe: Sprinkle the filings and sketch the actual pattern.

Question 4. Compare and Reflect:

1. Where does your prediction match the observed pattern, and where does it differ?
2. Is the field pattern in this activity ever simply the two single-dipole patterns added side by side with no distortion? Why or why not?

Question 5. A Line of Stacked Magnets

Stack several magnets with poles aligned (N up on every magnets) to form a short column, and repeat the iron-filings observation.

1. Examine the field pattern near one end of the stack. How does it compare to the single-magnet pattern from Part A?
2. Now examine the pattern along the side of the stack, away from either end. How is this pattern different from what you saw near the end?

Sketch both regions. You will return to this pattern later in the session — keep your sketch.

Question 6. Reflection

Why are magnetic field patterns sensitive to orientation, while gravity fields are not?

Key Ideas

Magnetic fields are dipolar, not monopolar. Every magnet has both a north and south pole; the field lines form closed loops. This gives magnetic anomalies a more complex shape than gravity anomalies, which arise from monopole-like mass distributions.

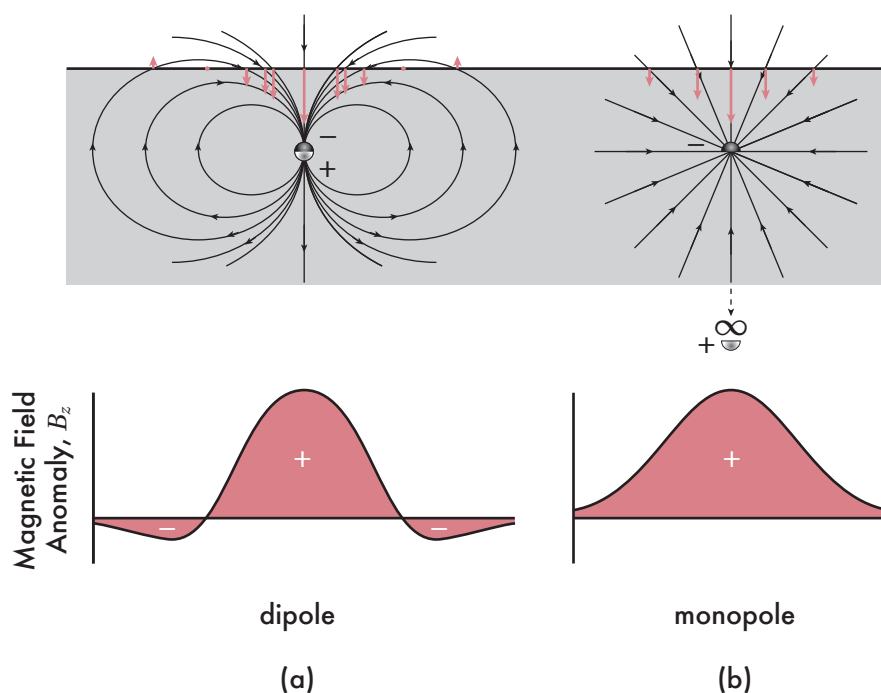


Figure 1: The (b) monopolar field of a point mass and the (a) dipolar field of a magnetic point source.

Orientation controls anomaly shape. Rotating the same source changes the pattern of positive and negative lobes without changing the source volume or susceptibility. This is the fundamental reason why magnetic interpretation requires knowledge of the inducing field direction and source magnetization direction.

A line of aligned dipoles can look like a single pole — but only up close. Stacking magnets pole-to-pole cancels every internal north-south junction, leaving only the two outer poles, now separated by the full length of the stack. Close to one end, that far pole is negligible and the field looks like a single isolated pole. Far from the stack, both poles matter again and the field must revert to a two-lobed, dipolar pattern — because total magnetic charge is always exactly zero, no matter how a source is shaped or arranged. This is the same limit you tested with the cube magnet: no geometry escapes it, only postpones it to shorter range.

Activity 6.B

Magnetic Structures

Purpose

This activity mirrors previous gravity workshops. You will reason about magnetic anomaly shapes and compare them directly with gravity anomalies from the same structures.

Learning Outcomes

By the end of this activity, you should be able to:

- Sketch the expected vertical magnetic anomaly for a range of simple geological structures.
- Identify the sign, symmetry, width, location of steepest gradients, and asymptotic behavior for each structure.
- Compare magnetic and gravity anomalies for the same body and explain key differences in terms of the dipolar nature of magnetic fields.
- Explain why magnetic interpretation is more non-unique than gravity interpretation.

Assume:

- Uniform susceptibility
- Induced magnetization only
- Constant inducing field

Background

Before sketching, it's worth anticipating one basic difference between gravity and magnetic fields from a compact source. Stripped down to the geometry alone — dropping mass, susceptibility, and other physical constants — the two fields scale with distance r from the source as

$$g \propto \frac{1}{r^2} \quad B \propto \frac{1}{r^3}.$$

A magnetic field falls off one power of r faster than the equivalent gravity field, because every magnetic source has both a positive and a negative pole rather than a single sign of mass. Practically, this means a magnetic anomaly from a compact, near-surface source should look *narrower and more sharply peaked* than a gravity anomaly from a source of the same size and depth. The same one-power-faster relationship holds for extended sources too, such as the cylinder and slabs below, even though the exact exponents shift with source geometry.

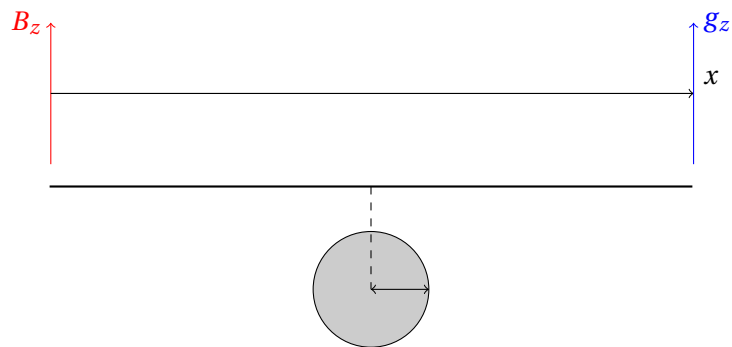
Instructions

For each structure:

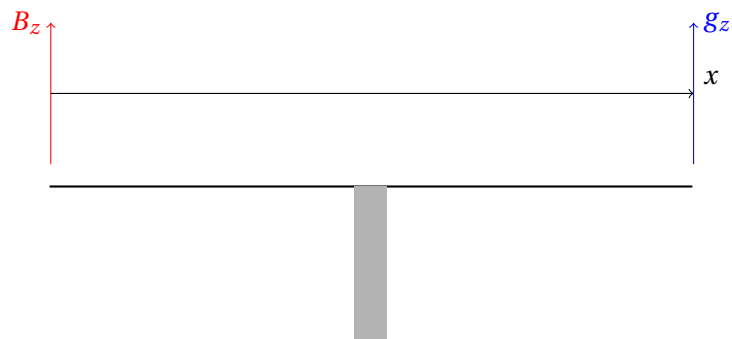
1. Draw the gravity anomaly
2. Sketch the expected vertical magnetic field associated with a vertical magnetization
3. Indicate sign, symmetry, width, and strongest gradient

Structures

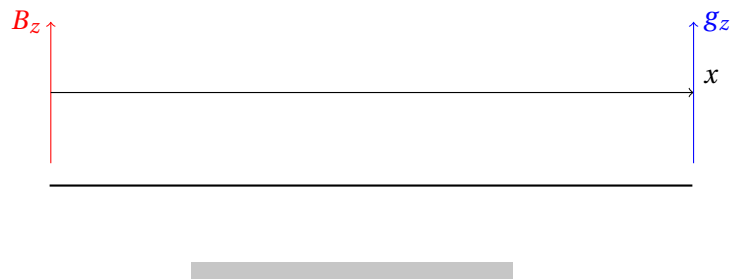
A) Buried Cylinder (2D; infinite strike)



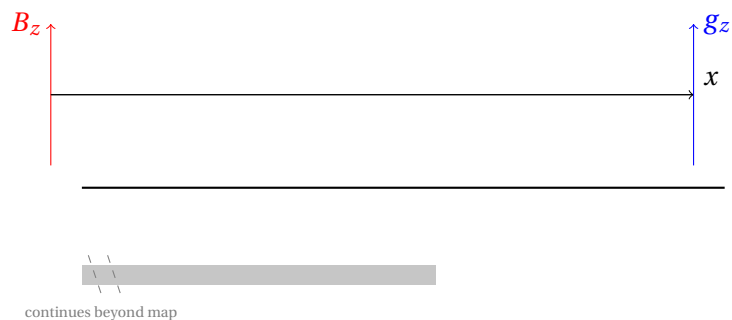
B) Vertical Slab



C) Finite-Extent Horizontal Slab



D) Semi-Infinite Horizontal Slab



Question 1. Vertical vs Horizontal Inducing Field

Repeat one case assuming a horizontal inducing field.

1. What are the major differences between the magnetic and gravity anomalies?

2. How would the anomaly for the sphere and finite horizontal slab change if the magnetization were horizontal rather than vertically oriented? Remember, we are still looking at B_z .

3. How do the differences between gravity and magnetic anomalies reduce the non-uniqueness of geologic interpretations?

Question 2. Reflection

Give two reasons why magnetic interpretation is more non-unique than gravity.

Key Ideas

Sign and shape depend on source geometry and field direction, not just on source strength. A mass excess always produces a positive gravity anomaly, full stop. A susceptibility contrast produces a magnetic anomaly whose sign and shape depend on the body's geometry *and* on the direction of the inducing field — at the magnetic pole a symmetric body gives a symmetric, bipolar anomaly; at lower latitudes the same body gives an asymmetric one with shifted lobes. Gravity has no equivalent to either effect. This traces back to the same monopole-versus-dipole distinction that gives magnetic fields their more complex geometry: a mass excess has only one sign, but every magnetic source carries both a north and a south pole.

Finite structures produce edge effects the end-member sketches don't show. Near the single edge of Structure D, the anomaly peaks just inside the edge and dips to a trough just outside it, then returns fully to zero further away — a point near the edge sits closer to source material on one side than the other, the same near-vs-far dipole reasoning behind the $1/r^3$ falloff above, just applied across an extended source. Structure C has *two* such edges. If they're far enough apart, each produces its own peak-and-trough and the interior

settles to a shallow minimum rather than true zero; if they're close together, the two signatures can merge into something that looks more like a single smooth bump. Compare your sketches for C and D directly — can you tell which regime C is in?

Non-uniqueness is greater in magnetism than gravity. Three unknowns — susceptibility, geometry, and magnetization direction — must all be constrained simultaneously. Remanent magnetization (a frozen-in field direction different from today's) adds a fourth unknown. Additional constraints from geology, paleomagnetism, or other data are usually required to discriminate between models.

Activity 6.C

Magnetic Susceptibility of Rocks

Purpose

This activity links magnetic susceptibility to mineralogy and geological processes such as alteration, metamorphism, weathering, and igneous differentiation.

Learning Outcomes

By the end of this activity, you should be able to:

- Measure magnetic susceptibility on a range of rock and ore samples and record observations systematically.
- Relate susceptibility values to mineralogy, particularly the role of magnetite and other iron oxides.
- Distinguish primary (igneous) from secondary (alteration or metamorphic) controls on rock magnetic susceptibility.
- Discuss why susceptibility is not a simple proxy for bulk rock composition or metal content.

Background

Magnetic susceptibility (χ) describes how strongly a material becomes magnetized in response to a field:

$$\mathbf{M} = \chi \mathbf{H}.$$

Unlike density, which depends on the bulk mineralogy of a rock, susceptibility is controlled almost entirely by a trace phase. Magnetite's susceptibility is orders of magnitude higher than that of the silicate minerals that make up the bulk of most rocks, so even a few percent by volume of magnetite dominates the measured susceptibility regardless of what the rest of the rock is made of. A handful of Ni- and Co-bearing minerals can also contribute, but at typical crustal abundances their effect is negligible next to magnetite's.

Because susceptibility is set by a trace phase, it is sensitive to exactly which Fe-Ti oxide is present, not simply how much iron the rock contains. Figure ?? shows the relevant Fe-Ti oxide system: magnetite and its Ti-substituted relatives at one end, hematite and ilmenite at the other. Today's samples will let you see this system in action — the same starting rock can end up on either side of it, depending on what reaction it underwent.

Samples and Observations

Station 1: Reaction Pairs

Question 1. Reaction Pairs

1. Compare the peridotite and serpentinite susceptibility. How does serpentinization lead to high magnetic susceptibility?

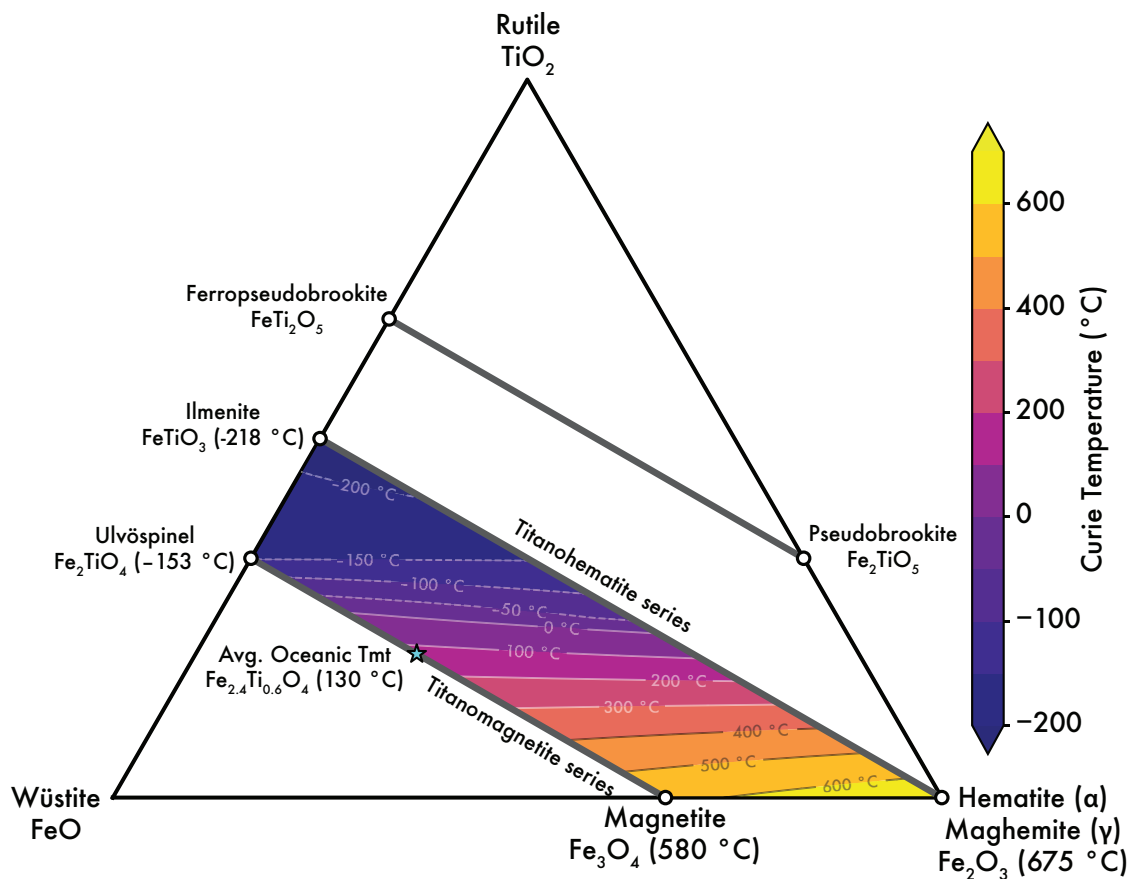


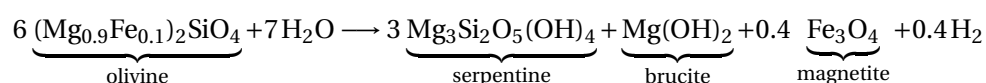
Figure 2: The Fe-Ti oxide system. Which side of this diagram a rock's oxide phase falls on — not its bulk iron content — determines whether it is magnetic.

2. Compare the fresh basalt and altered seafloor basalt. Why can alteration dramatically reduce susceptibility in mafic rocks?

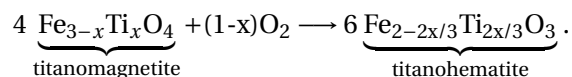
3. Compare the granite or sedimentary protolith with its metamorphosed equivalent. Which minerals control susceptibility in felsic or sedimentary rocks?

Sample	Rock type	Geological context	Susceptibility (SI)	Notes
1	peridotite			
2	serpentinite			
3	basalt (fresh)			
4	basalt (altered)			
5	eclogite			

Serpentinization of olivine is commonly associated with reactions such as



while oxidative weathering of basalt may involve reactions such as



Question 2. Cross-Comparison: Two Paths to Low Susceptibility

Both the altered seafloor basalt and the eclogite are weakly magnetic, but they reached that state by different geological histories. Propose one observation (mineralogical, textural, or geochemical) that could distinguish which process — surface alteration or deep metamorphism — produced a given low-susceptibility sample.

Question 3. Cross-Comparison: One Process, Two Outcomes

Hydrothermal alteration acted on both the peridotite and the basalt in this activity, but with opposite magnetic consequences: serpentinization of peridotite *produces* magnetite, while seawater alteration of basalt *destroys* it. Given that both are altered by broadly similar fluid-rock interaction, what property of the starting rock — rather than the process itself — determines whether magnetite is created or consumed?

Station 2: Igneous Suite

Sample	Rock type	Predicted rank	Susceptibility (SI)	Notes
6	gabbro			
7	diorite			
8	granodiorite			
9	granite 1			
10	granite 2			

Question 4. Igneous Suite

- Before measuring, rank the igneous samples from most to least magnetic based on their appearance (color index, inferred mafic mineral content).

- After measuring, compare your predicted ranking to the results. Does susceptibility track color index as expected? What might cause a sample to depart from the trend?

Station 3: Ore Minerals

Sample	Mineral	Predicted rank	Susceptibility (SI)	Notes
11	magnetite			
12	galena			
13	tin ore			

Question 5. Ore Minerals

- Before measuring, rank the four ore samples from most to least magnetic, based on their appearance and metal content.

2. After measuring, compare your prediction to the results. Which sample most surprised you, and why?

3. Pyrite contains iron. Why is it only weakly magnetic, if at all, despite this?

Complete at the end

Question 6. Synthesis

1. Why is susceptibility not a simple proxy for bulk composition or metal content?

2. Which geological processes and settings tend to increase susceptibility, and which tend to decrease it?

Question 7. Reflection

Explain why petrology is essential for interpreting magnetic anomalies.

Key Ideas

Magnetite controls crustal magnetization. Because the susceptibility of magnetite is orders of magnitude higher than that of common silicate minerals, even a few percent magnetite can dominate the magnetic response of a rock. Susceptibility therefore reflects the abundance of magnetic Fe-Ti oxides rather than bulk composition or metal content. In igneous rocks, this abundance commonly decreases as melts evolve toward more felsic compositions, producing systematic variations in susceptibility across an igneous suite through primary crystallization alone. No alteration or metamorphic overprint is required to generate these magnetic contrasts.

Sorry physics isn't the answer to everything. Petrologic history is essential for magnetic interpretation. Without knowing the mineralogy, alteration history, and redox evolution of the source rocks, magnetic anomalies can be highly ambiguous. The same bulk composition may produce very different magnetic responses depending on the abundance and stability of magnetic minerals such as magnetite or titanomagnetite. Likewise, processes such as alteration and metamorphism do not have a fixed effect on susceptibility: depending on the initial iron content, oxidation state, and fluid or melt chemistry, they may either create or destroy magnetic minerals. Consequently, interpreting magnetic anomalies requires an understanding of the underlying reaction geochemistry and mineralogical evolution, not simply the identification of a geological process.

Activity 6.D

Curie Depth and Magnetic Crustal Thickness

Purpose

This activity explores how temperature limits crustal magnetization and why deep magnetic structure is long-wavelength.

Learning Outcomes

By the end of this activity, you should be able to:

- Explain how the Curie temperature limits the depth of crustal magnetization.
- Relate anomaly wavelength to source depth using the depth–wavelength principle.
- Compare seismic and magnetic estimates of crustal thickness and explain why they can differ.
- Estimate heat flow from Curie depth using a simple geothermal gradient.

Question 1. Curie Depth Concept

1. What mineral dominates crustal magnetization?

2. What happens above the Curie temperature?

3. Is Curie depth a thermal or compositional boundary?

Question 2. Depth and Wavelength

Given several magnetic profiles:

1. Which correspond to shallow sources?

2. Which to deep Curie depth?

3. How does depth affect short wavelengths?

Question 3. Antarctica Case Study

Consider the magnetic and crustal structure of Antarctica (Fig. ??).

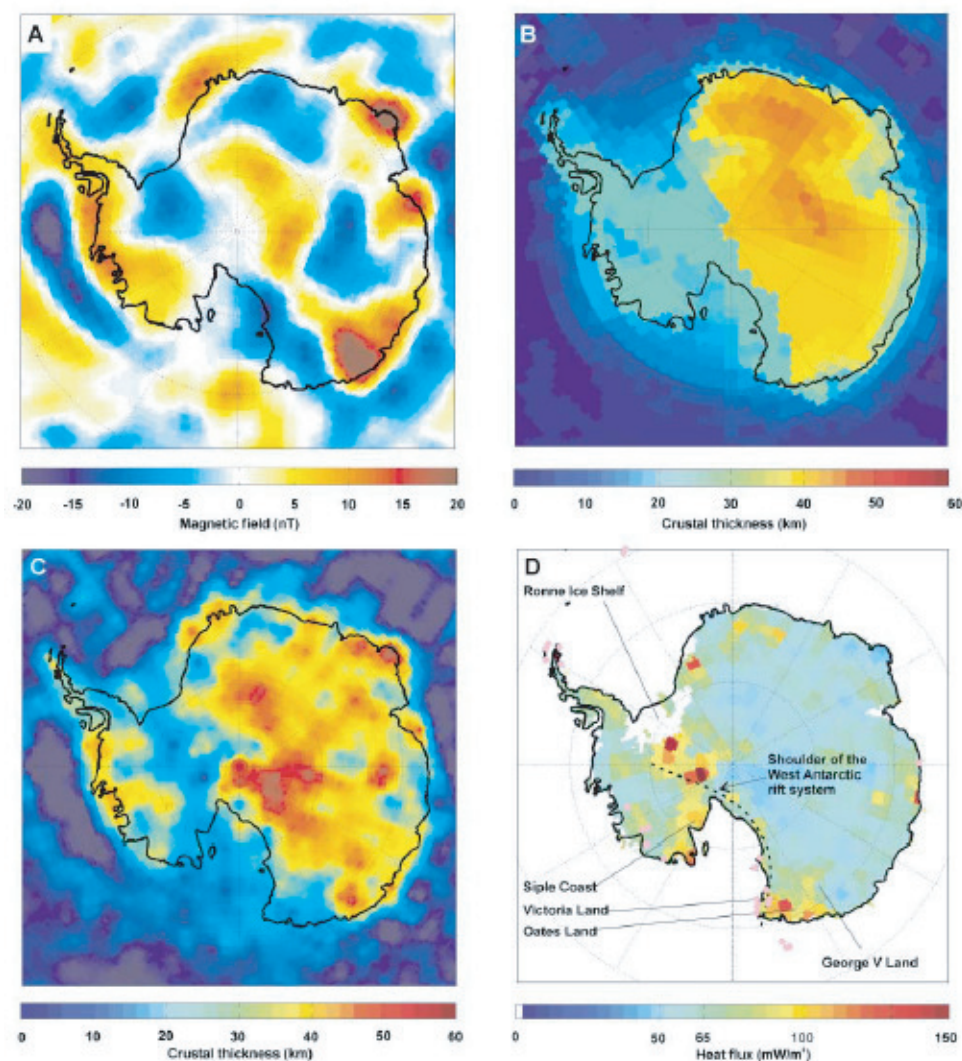


Figure 3: Estimating the Curie depth from vertical magnetic data. (a) Vertical magnetic field. (b) Crustal thickness model of Antarctica. (c) Estimate to base of magnetized layer. (d) Estimated heat flow. Figure from Maule et al. (**Science**, 2005).

Compare:

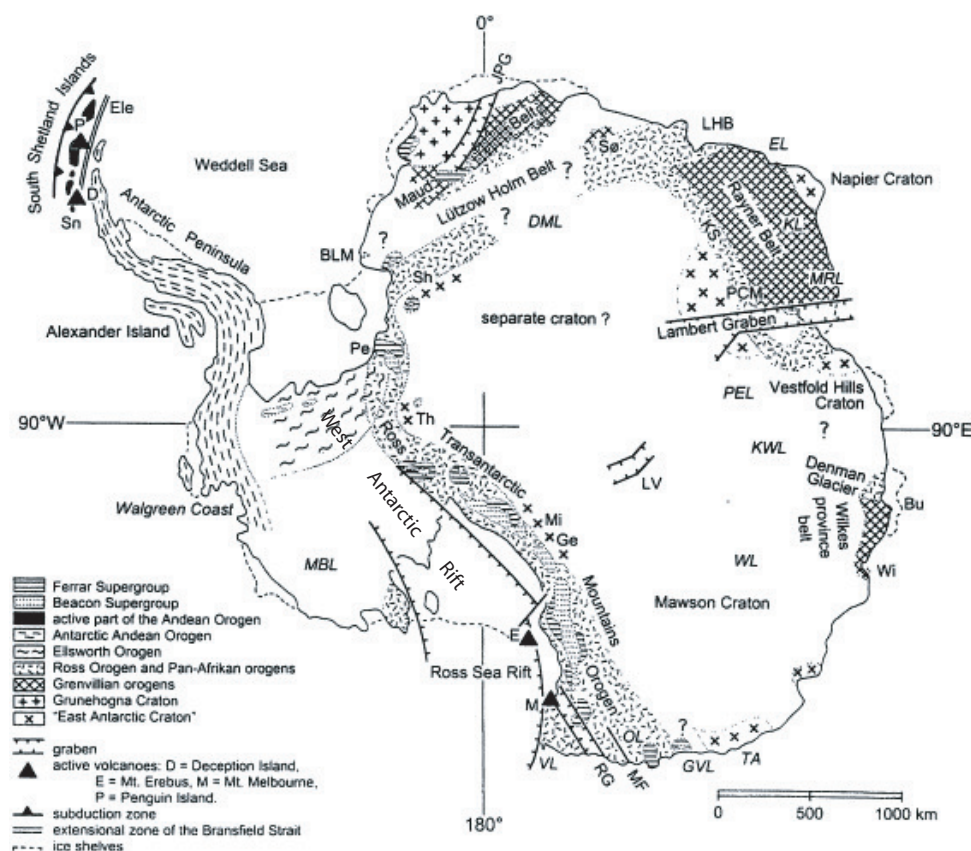
- Magnetic anomalies

- Seismic crustal thickness
- Magnetic crustal thickness

Why do seismic and magnetic thickness estimates differ?



The first-order geology of Antarctica is characterized by a stable (cratonic) region in East Antarctica and a Cenozoic rift in West Antarctica (Fig. ??).



Talarico and Kleinschmidt, in **Antarctic Climate Evolution**, 2008

Figure 4: Tectonic map of Antarctica from Talarico and Kleinschmidt (**Antarctic Climate Evolution**, 2008).

Question 4. Heat-Flow Estimate

The relationship below follows from Fourier's law of steady-state heat conduction, assuming a linear temperature increase from the surface to the Curie isotherm. We'll discuss this properly when we cover conductive heat flow in Week 8 — for now, take it as given and use it to estimate heat flow from the Curie depths you found in Part C.

Assume:

$$T_C = 580^\circ\text{C}, \quad k = 2.8 \text{ W m}^{-1} \text{ K}^{-1}$$

$$q = k \frac{T_s - T_C}{z}$$

where T_s is the surface temperature in $^\circ\text{C}$.

Estimate heat flow for a rift and a craton.

Question 5. Reflection

Explain why Curie depth makes magnetics sensitive to thermal structure.

Key Ideas

The Curie temperature defines the base of crustal magnetization. Above this temperature (about 580°C for magnetite), thermal energy prevents the alignment of magnetic domains and the rock is effectively non-magnetic. The depth at which crustal temperatures reach the Curie point therefore sets the maximum depth from which magnetic anomalies can originate.

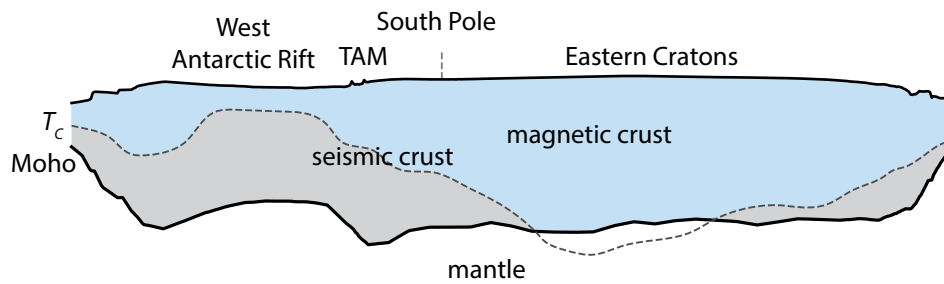


Figure 5: The magnetic crustal thickness differs from the seismic crust, which is defined by the depth to the Moho. The magnetic crustal thickness however is defined by the depth at which rocks are no longer magnetic. This depth is typically defined by the Curie Isotherm, T_C (approximate closure temperature of magnetite). There are places where the isotherm dips below the Moho, into the mantle which typically has very low susceptibility. In such cases, the magnetic crustal thickness is compositionally rather than thermally controlled. While this is a real cross section of crustal thickness and topography of Antarctica, the Curie isotherm is hypothetical.

Magnetic and seismic Moho usually differ, and for a specific reason. Because Curie depth tracks the geotherm rather than composition, a hot, thin crust (e.g., a rift zone) can reach T_C within the crust, while a cold, thick craton can push T_C down to, or below, the seismic Moho. Where the isotherm sits above the Moho, the mismatch is thermally controlled; where it would sit below the Moho, the mantle's low susceptibility takes over and the boundary becomes compositional instead. Magnetic crustal thickness is therefore not a direct proxy for seismic crustal thickness—which one controls it depends on tectonic setting.

Curie depth provides a heat-flow estimate. Assuming a linear geothermal gradient and a known Curie temperature, $q \approx kT_C/z_C$ gives a first-order heat-flow estimate. This is a powerful application because it provides continental-scale heat-flow maps where borehole measurements are unavailable (e.g., beneath Antarctic ice).